COM 3401–
DATA COMMUNICATION AND NETWORKING

Lecture 05

DATA TRANSMISSION

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Topics covered

- Introduction
- Data and Signals
- Digital Transmission
- Analog Transmission
- Transmission Modes
- Transmission Impairments

Reference: William Stallings, *Data and Computer Communications*, 8th Edition

5.1 Introduction



What is Data Transmission?

- Data transmission occurs between:
  - A transmitter
  - A receiver
- Involves a transmission medium

- Key idea:
  - Information must be represented by signals suitable for transmission.
  - Communication takes place through the propagation and processing of signals.

- The successful transmission of data depends principally on:
  - The quality of the signal
  - The characteristics of the transmission medium

Data, Signals, Signaling and Transmission

Term	Meaning
Data	Entities that convey meaning or information
Signal	Electrical or electromagnetic representation of data
Signaling	Physical propagation of the signal along a suitable medium
Transmission	Communication of data through the propagation and processing of signals

The action flow:

Data → Signal → Transmission Medium → Received Signal → Data

Analog and Digital: The Basic Distinction

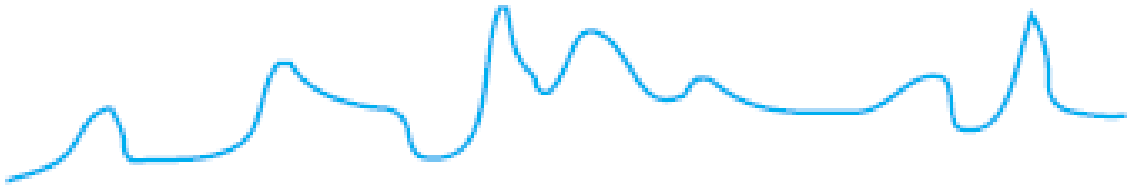
- The terms analog and digital correspond roughly to:
  - Analog → Continuous
  - Digital → Discrete
- They can be applied to:
  1. Data
  2. Signals
  3. Transmission

- Do not equate:
  - Analog = old technology
  - Digital = modern technology
- The distinction is about the nature of the data/signal, not simply whether the technology is old or new.

Analog Data

- Take on continuous values within an interval.
- Change smoothly rather than taking only discrete values.
- Examples :
  - Voice
  - Temperature
  - Pressure



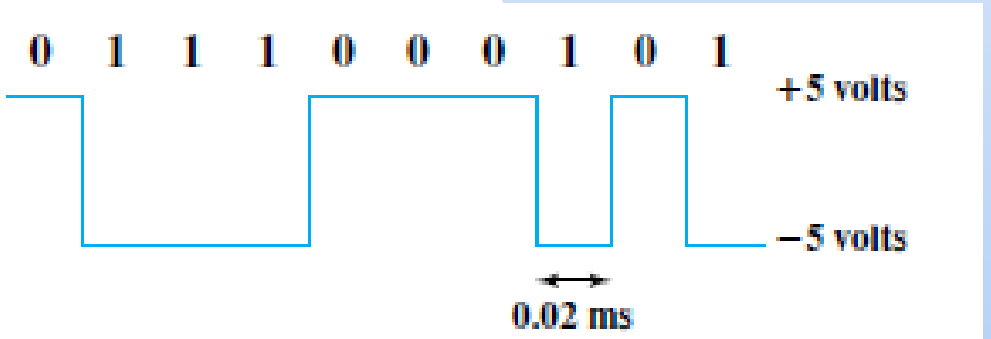


- Voice:
  - Acoustic sound waves represent analog data.
  - Speech consists of continuously varying patterns of intensity.

Digital Data

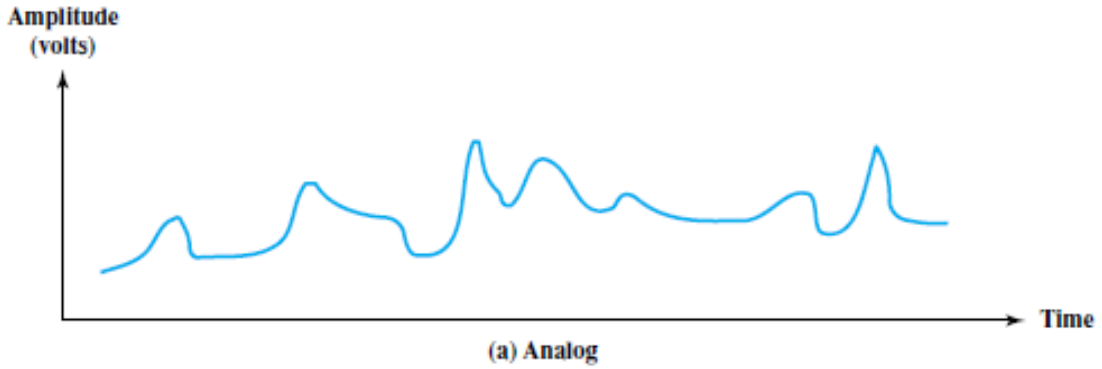
- Take on discrete values.
- Do not vary continuously through every possible value.
- Examples:
  - Text
  - Integers
  - Binary information



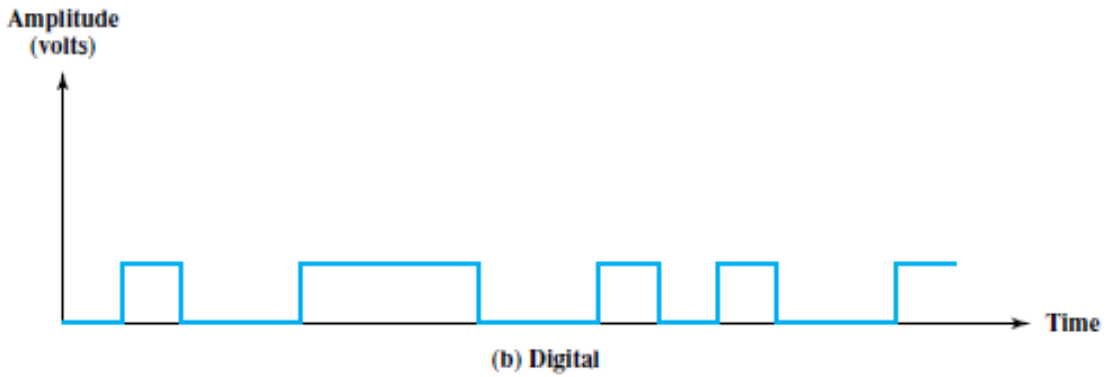


Analog Signal vs Digital Signal

Aspect	Analog Signal	Digital Signal
Signal intensity	Signal intensity varies smoothly over time.	Signal intensity maintains a constant level for some period.
Nature of the wave	No breaks or discontinuities in the ideal model.	It then changes abruptly to another constant level.



(a) Analog



(b) Digital

Periodic and Aperiodic Signals

Periodic Signal

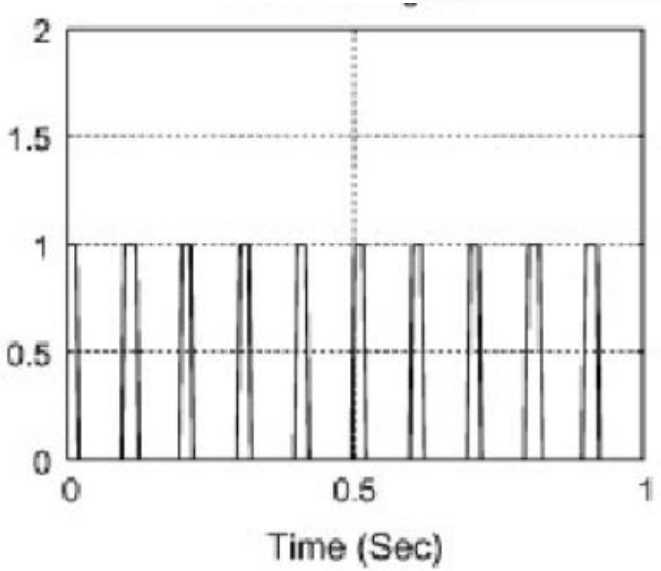
- A signal is periodic when the same signal pattern repeats over time.
- For a periodic signal:

$$s(t + T) = s(t)$$

where:

T = period

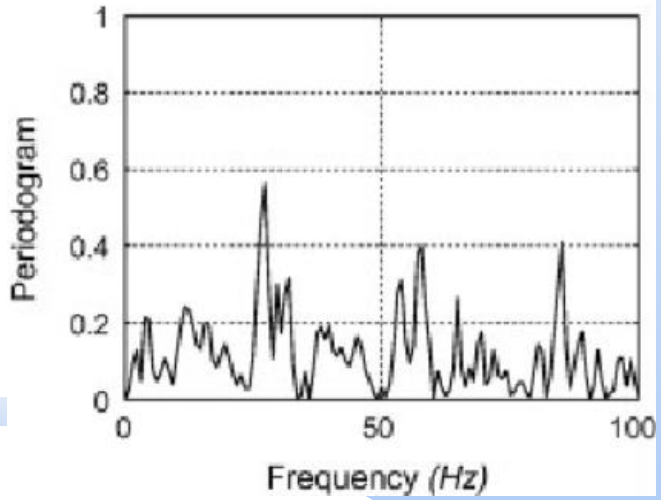
- The period is the smallest value for which the signal repeats.
  - E.g.: Sine wave



Periodic Signal

Aperiodic Signal

- Does not repeat in a regular pattern.



Aperiodic Signal

Characteristics of a Sine Wave

1. Peak Amplitude (A)

- Maximum strength of the signal.
- Typically measured in volts.

2. Frequency (f)

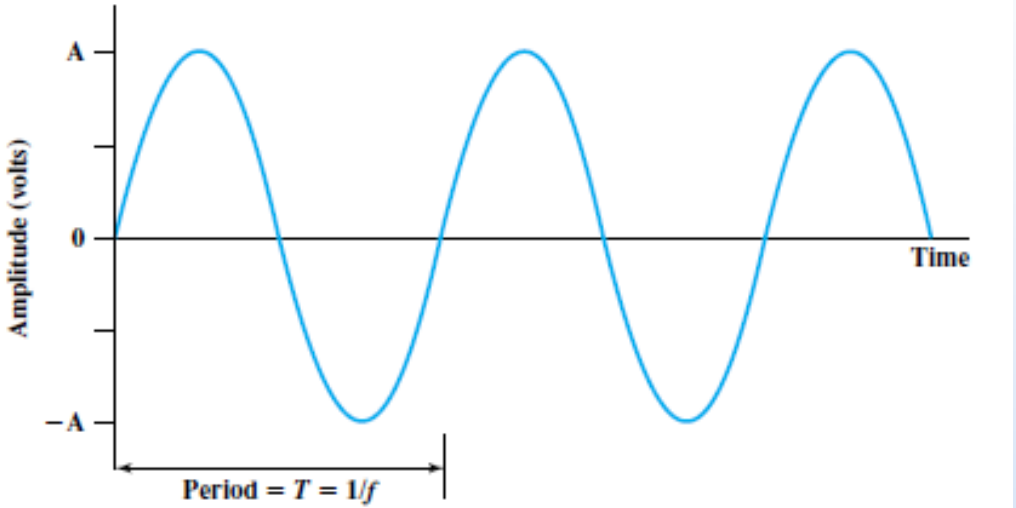
- Rate at which the signal repeats.
- Measured in Hertz (Hz).

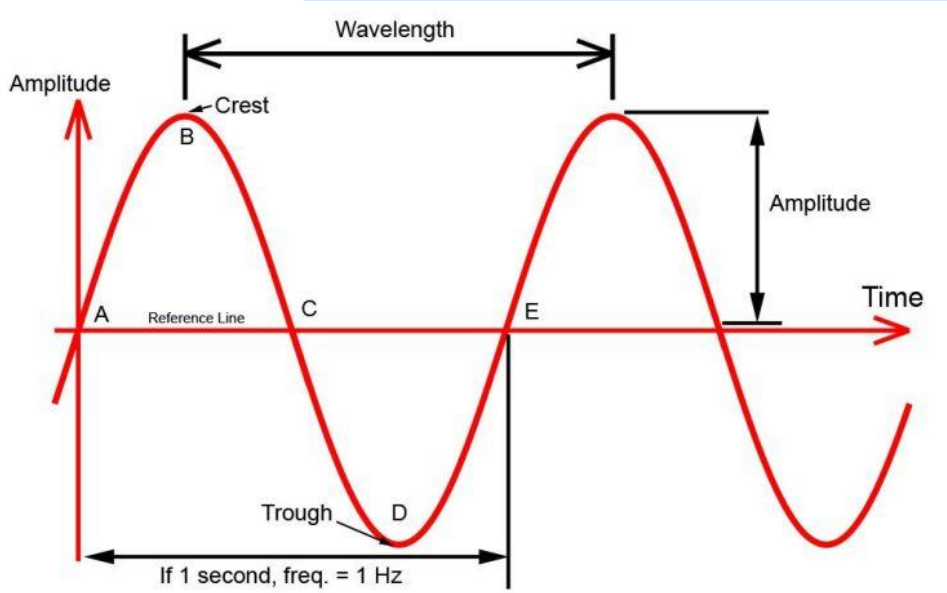
3. Phase (φ)

- Relative position of the signal in time.

4. Wavelength (λ)

- Distance occupied by a single cycle





Period and Frequency

Period

- The period T is the time required for one complete cycle.

$$T = \frac{1}{f}$$

and:

$$f = \frac{1}{T}$$

Where:

T = period

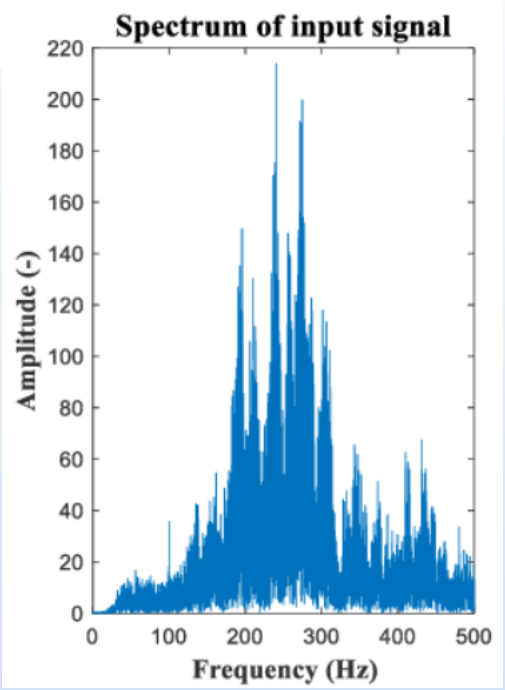
f = frequency

- Example:

If: $f = 1\,Hz$

then: T=1second

Frequency, Spectrum and Bandwidth



Spectrum

- The spectrum of a signal is the range of frequencies that it contains.

Absolute Bandwidth

- The width of the signal's spectrum.

$$Bandwidth = f_{highest} - f_{lowest}$$

Effective Bandwidth

- Most practical signals may contain a broad range of frequencies.
- However, most of the signal energy may be concentrated within a narrower frequency range.
- This range is referred to as the effective bandwidth, or simply bandwidth.

Bandwidth and Data Rate

- **Bandwidth:** The maximum amount of data that a communication channel can carry in a given period of time. It is usually measured in bits per second (bps), such as Mbps or Gbps.
- **Data Rate:** The actual amount of data transmitted through a communication channel per second. It is also measured in bits per second (bps).

- An important observation:
  - Higher data rate → greater required effective bandwidth.
- Conversely:
  - Greater transmission-system bandwidth → potentially higher data rate.

- But the actual data rate is also affected by:
  - Transmission impairments
  - Acceptable error rate

5.2
Data and Signals



Data and Signals

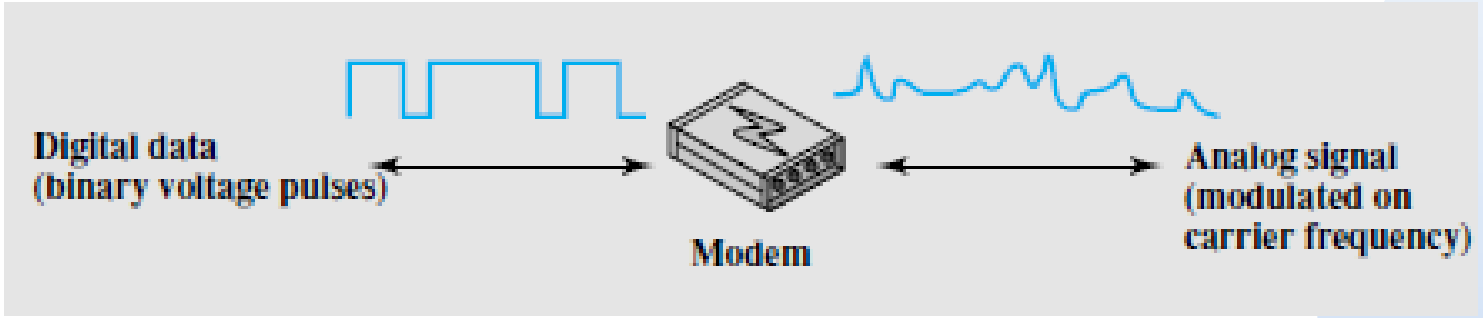
- Both analog and digital data can be represented by either analog or digital signals.

	Analog Signal	**Digital Signal**
Analog Data	Analog data → Analog signal • *Voice represented by an analog signal*	Analog data → Digital signal • *Voice converted into a bit stream using a codec*
Digital Data	Digital data → Analog signal • *Digital data encoded onto a carrier using a modem*	Digital data → Digital signal • *Binary data represented by voltage pulses*

Digital Data → Analog Signal

Using a Modem

- A modem converts digital data into an analog signal.



- Process:

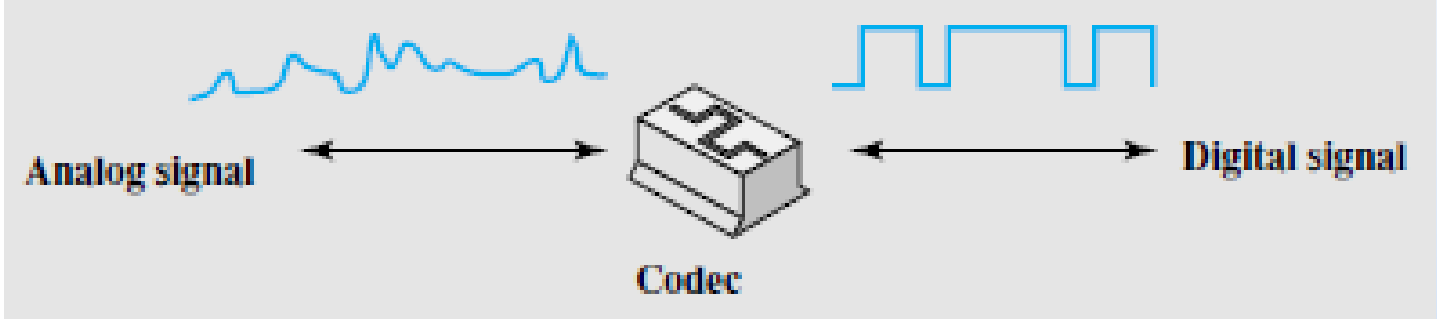
Digital data ➡️ Binary voltage pulses ➡️ Modem ➡️ Analog signal

- The modem encodes digital data onto a carrier frequency.
- At the receiving end:

Analog signal ➡️ Modem ➡️ Digital data

Analog Data → Digital Signal

- Using a Codec
- A codec (coder-decoder) can convert analog data into digital form.



- For voice:

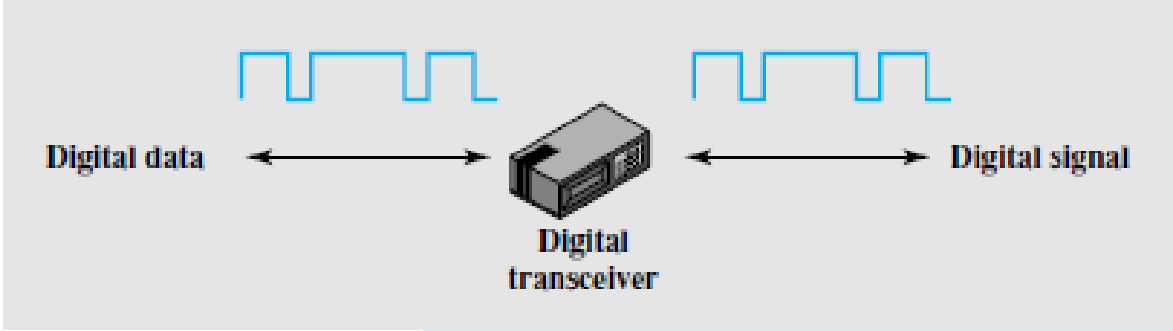
Analog voice ➡ Codec ➡ Bit stream

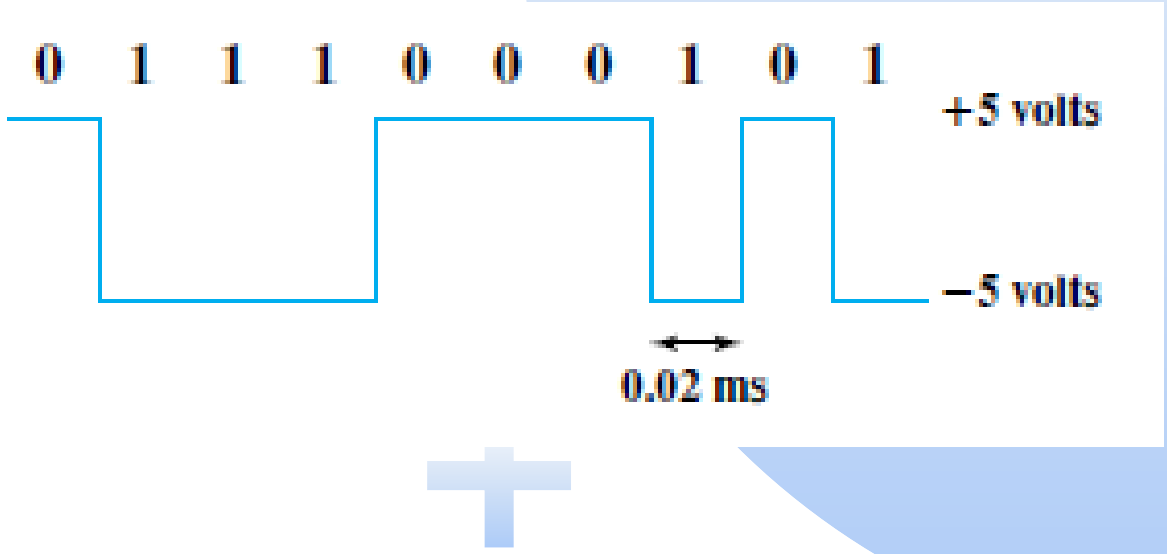
- At the receiver:

Bit stream ➡ Codec ➡ Reconstructed analog data

Digital Data → Digital Signal

- Digital data can be represented as a digital signal.
- A simple method:
  - Binary 1 → one voltage level
  - Binary 0 → another voltage level
- Example:
  - Binary 1 → -5 V
  - Binary 0 → +5 V
  - Each bit duration = 0.02 ms
  - Data rate = 50 kbps



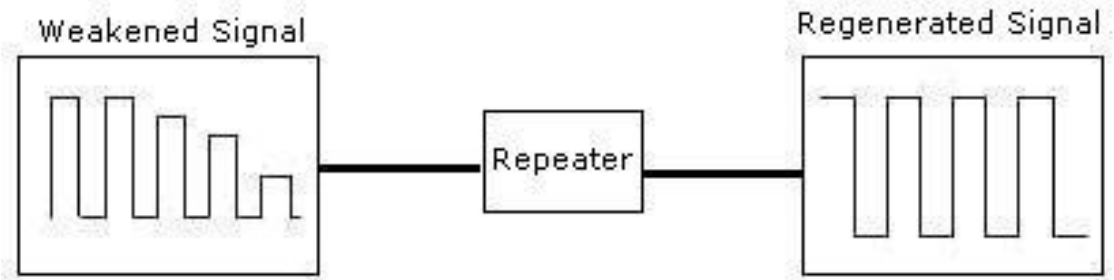


5.3
Digital Transmission



Digital Transmission

- Digital transmission assumes a binary content to the signal.
- A digital signal can travel only a limited distance before:
  - Attenuation
  - Noise
  - Other impairments
- threaten the integrity of the data.
- For longer distances, repeaters are used.

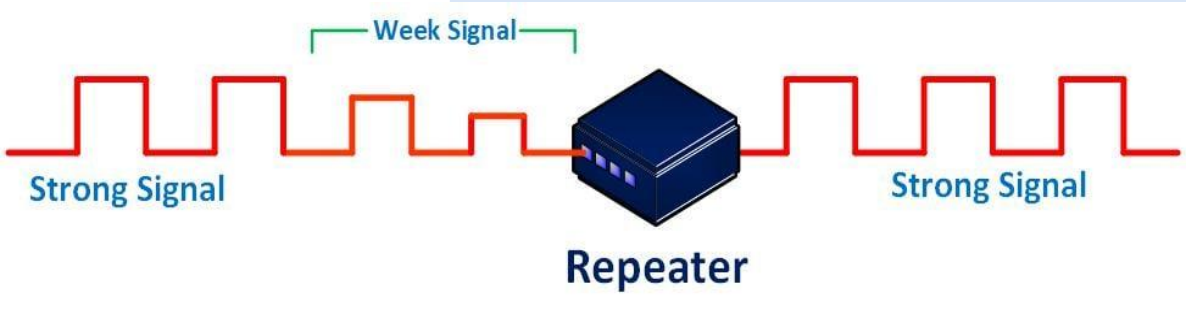


Sender ➡ Digital Signal ➡ Repeater ➡ Digital Signal ➡ Receiver

Digital Transmission: Repeaters

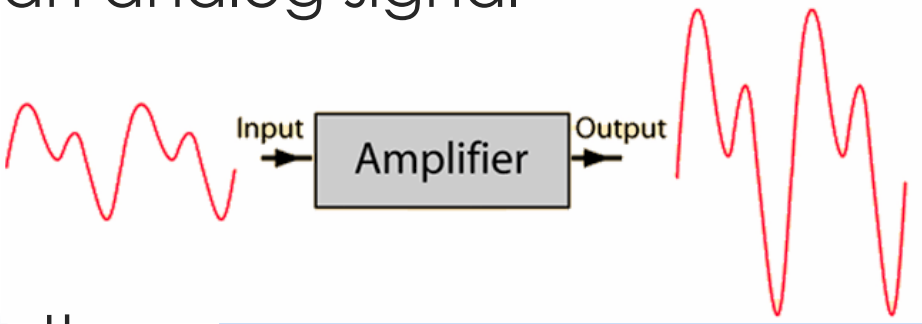
Repeater Operation

- A repeater:
  1. Receives the incoming digital signal.
  2. Recovers the pattern of 1s and 0s.
  3. Generates a new digital signal.
  4. Retransmits the new signal.
- Therefore:
  - Signal degradation does not simply accumulate in the same way as with cascaded amplifiers.



Analog Transmission

- Transmits an analog signal without regard to its content.
- The signal may represent:
  - Analog data, such as voice
  - Digital data that has been converted into an analog signal
- As the signal travels:
  - It becomes weaker due to attenuation.
  - Amplifiers are used to increase signal strength.



Source ➡️ Analog signal ➡️ Amplifier ➡️ Analog signal ➡️ Receiver

Analog Transmission: The Problem with Amplifiers

- An amplifier increases the energy of the signal.
- However:
  - **Amplifier also amplifies noise.**
- Therefore, with multiple amplifiers:
- **Signal + Noise ➡️ Amplifier ➡️ Stronger Signal + Stronger Noise ➡️ Another Amplifier ➡️ Increasing distortion**
- Cascaded amplifiers cause the signal to become increasingly distorted because the noise components are also amplified.

Digital vs Analog Transmission

Analog Transmission	Digital Transmission
Uses amplifiers	Uses repeaters
Amplifier boosts signal	Repeater recovers 1s and 0s
Noise is amplified	New signal is generated
Noise/distortion can accumulate	Noise is not cumulative in the same way
Signal quality progressively degrades	Data integrity can be maintained over longer distances

Why Digital Transmission is Preferred

1. Digital Technology

- LSI and VLSI technology reduced the cost and size of digital circuitry.

2. Data Integrity

- Repeaters prevent noise and impairment effects from accumulating.
- Longer-distance transmission is possible while maintaining data integrity.

3. Capacity Utilization

- High-bandwidth links require efficient multiplexing.
- Digital time-division techniques can be more easily and cheaply implemented.

4. Security and Privacy

- Encryption can readily be applied to digital data.
- Analog data can also be digitized and then encrypted.

5. Integration

- Voice, video and digital data can all be treated digitally.
- This allows integration of different types of information.

Digital Does Not Always Mean Digital Signal

Remember:

- **Digital transmission ≠ necessarily digital signal**
- An **analog signal can also carry digital data**.
- For example:

Digital data ➡️ Modem ➡️ Analog signal ➡️ Transmission ➡️ Repeater ➡️ Digital data recovery

- The transmission system may use repeaters to recover the digital data from the incoming analog signal and generate a clean analog signal.

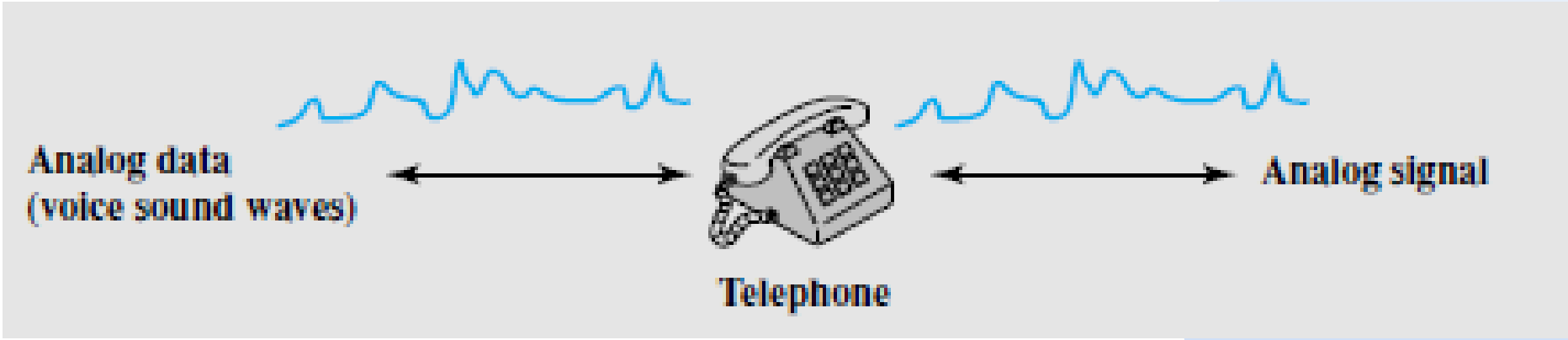
5.4
Analog Transmission



Analog Data and Analog Signal

Analog Data → Analog Signal

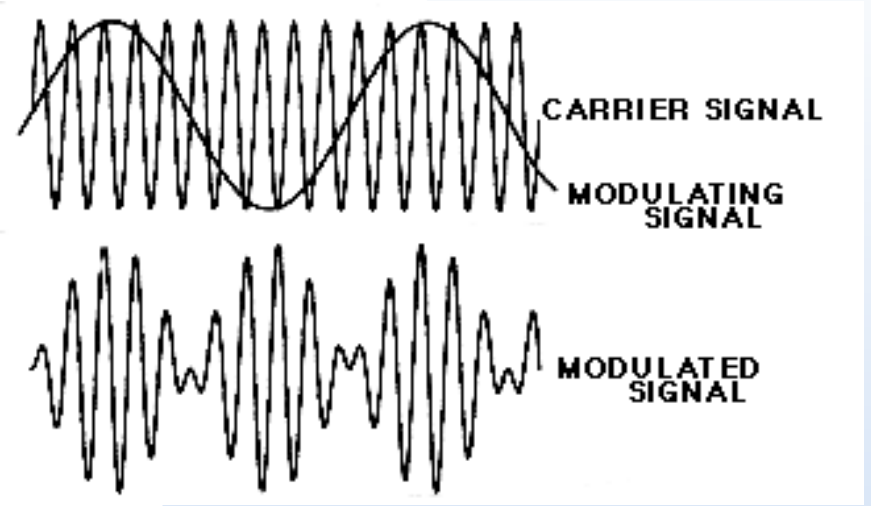
- Analog data such as voice can be represented by an analog signal.



- The signal can:
  - Occupy the same spectrum as the analog data, or
  - Be encoded to occupy another portion of the spectrum.

Digital Data and Analog Signal

Why use an analog signal for digital data?



- A modem converts:
  - Binary voltage pulses
  - Into an analog signal
  - By encoding the digital data onto a carrier frequency.
- The resulting analog signal occupies a spectrum centered around the carrier frequency.
- At the receiver, another modem **demodulates** the signal to recover the original data.

Analog Transmission vs Analog Signal

Be careful with terminology.

Analog Transmission

- Concerned with **how the signal is treated during transmission**.

Analog Signal

- Describes the **nature/form of the signal**.

Therefore:

- An analog signal can carry either analog data or digital data.

  And

- Analog transmission can be used for either type of data when the transmitted signal is analog.

5.5
Transmission Modes



Transmission Modes

There are three transmission modes based on the **direction of transmission between stations** :

1. Simplex
2. Half Duplex
3. Full Duplex

Simplex Transmission

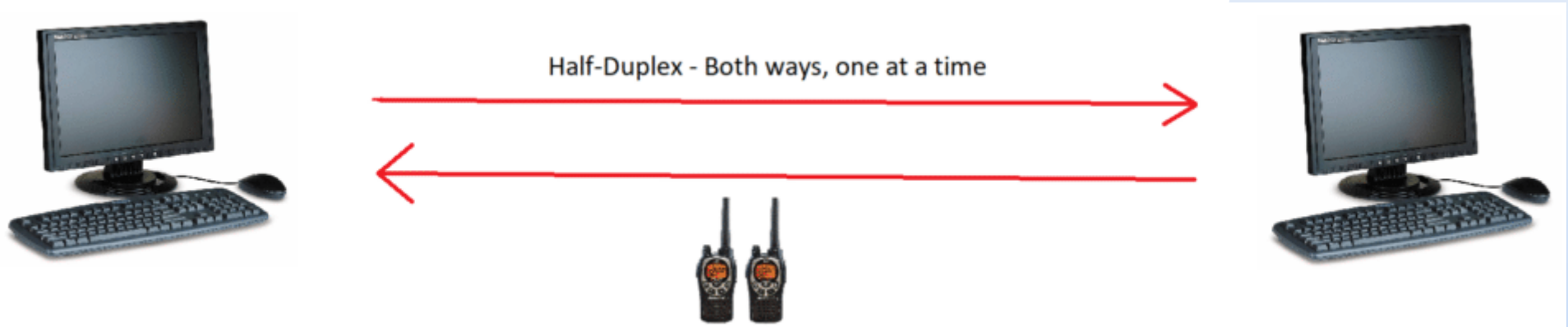
- Transmission occurs in only one direction.
- One station is the transmitter.
- The other station is the receiver.



- No return transmission.
- E.g.:
  - TV Remote control

Half-Duplex Transmission

- Both stations may transmit.
- Only one station transmits at a time.
- The stations take turns.



- No simultaneous transmission
- E.g.:
  - Walkie-Talkie

Full-Duplex Transmission

- Both stations may transmit simultaneously.
- Signals travel in both directions at the same time.



- E.g.:
  - Mobile phone

5.6 Transmission Impairments

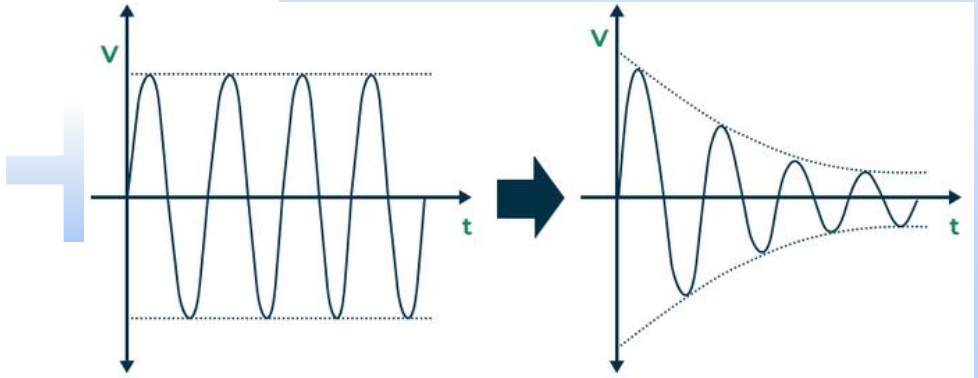


Transmission Impairments

- **What are Transmission Impairments?**
  - During transmission, the signal received at the destination may differ from the signal that was originally transmitted.
- Transmission impairments can:
  - Degrade the quality of analog signals
  - Introduce bit errors in digital signals
  - Reduce the information-carrying capability of a communication link
- Major Transmission Impairments
  - Attenuation and Attenuation Distortion
  - Delay Distortion
  - Noise

Attenuation

- **Attenuation** is the reduction in signal strength as a signal travels through a transmission medium.
  - Signal strength decreases with distance.
  - In guided media, attenuation is generally exponential.
  - It is typically expressed in decibels per unit distance.
  - In unguided media, attenuation depends more complexly on distance and atmospheric conditions.



Why is attenuation a problem?

- The received signal must:
  - Be strong enough for the receiver to detect.
  - Remain sufficiently above the noise level to avoid errors.

Compensating for Attenuation

How can attenuation be handled?

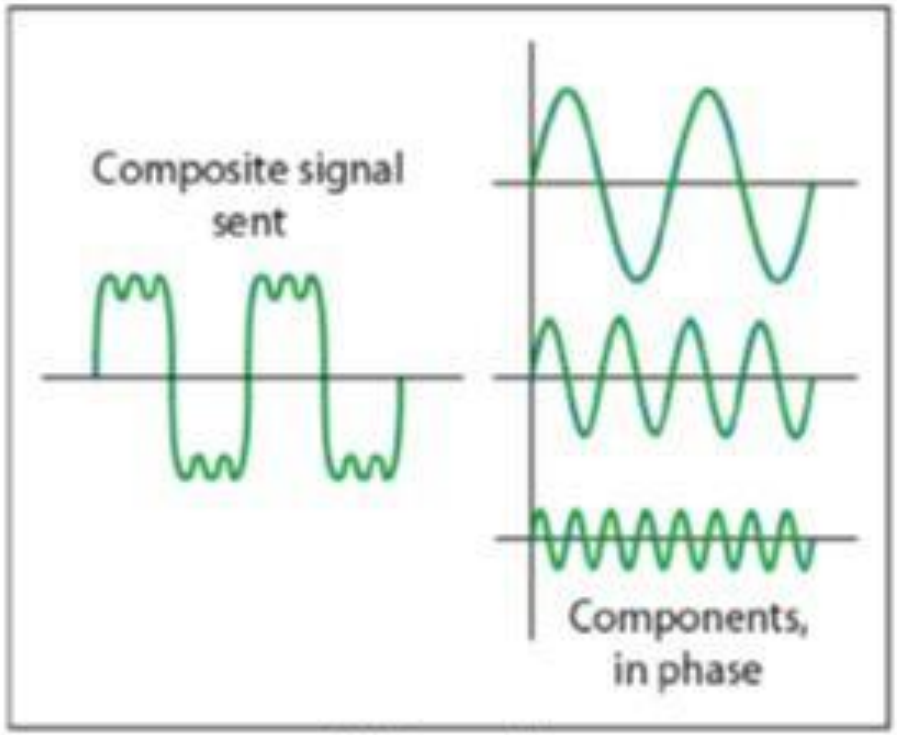
- Beyond a certain distance, attenuation becomes too great.
- Therefore:
  - Amplifiers or repeaters may be inserted at intervals.
  - Their purpose is to boost the signal.
- However, the signal cannot simply be made arbitrarily strong.
- If the signal is too strong, it may **overload transmitter or receiver circuitry**, causing distortion.

❖ **Signal must be strong enough to detect, but not so strong that it causes distortion.**

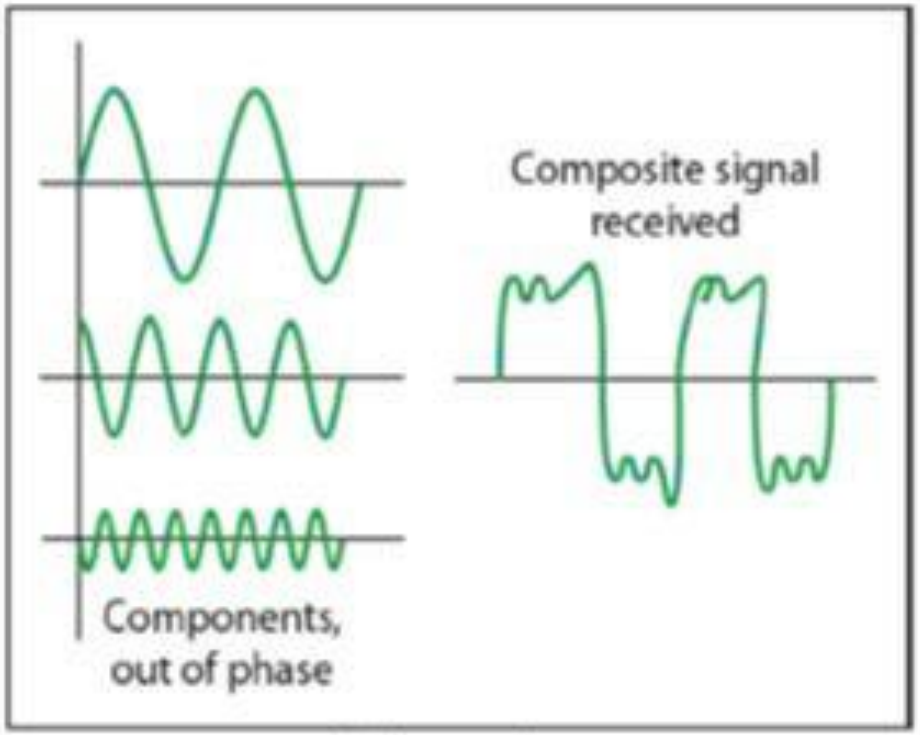
Attenuation Distortion

- **Attenuation is not necessarily the same at every frequency; Attenuation varies with frequency.**
- Therefore, different frequency components of a signal may be reduced by different amounts.
- This changes the shape of the received signal.
- **Particularly important for: Analog signals**
- For example, different frequency components of a voice signal may be attenuated differently, resulting in distortion and reduced intelligibility.

Attenuation Distortion



At the sender



At the receiver

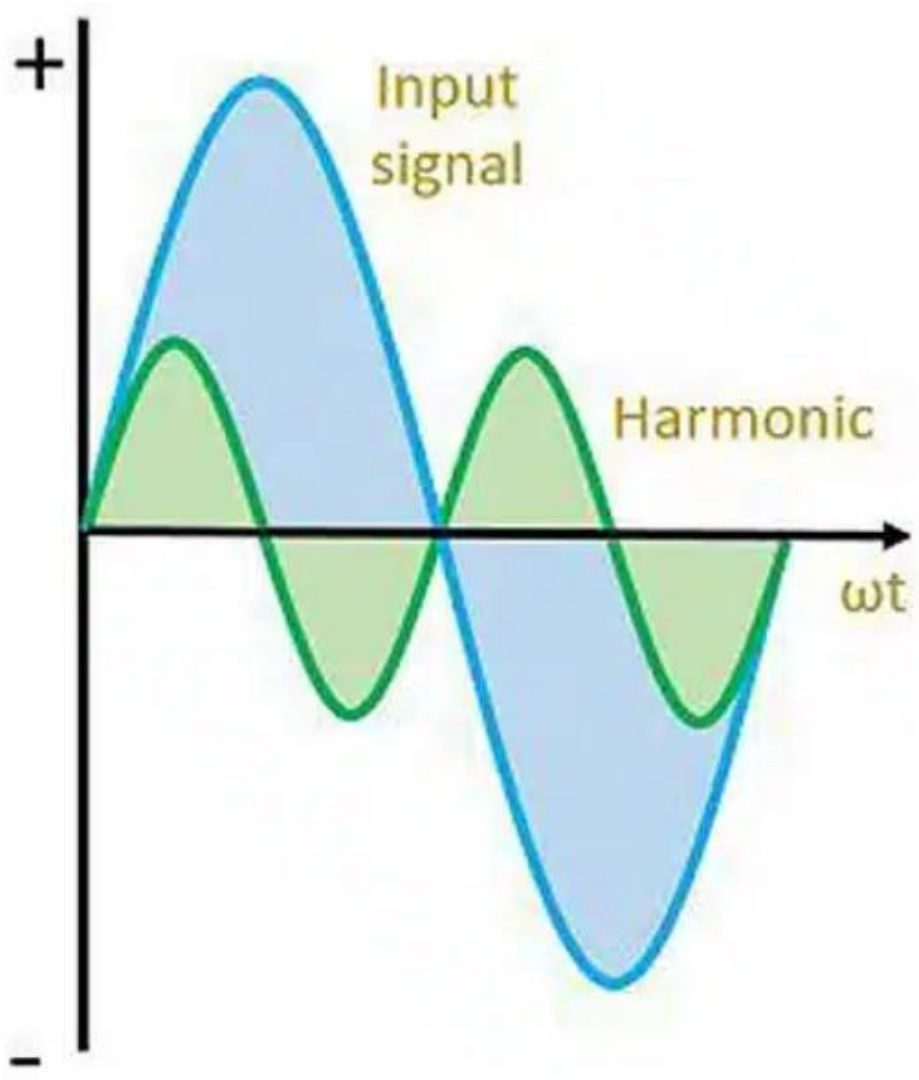
Equalization : Reducing Attenuation Distortion

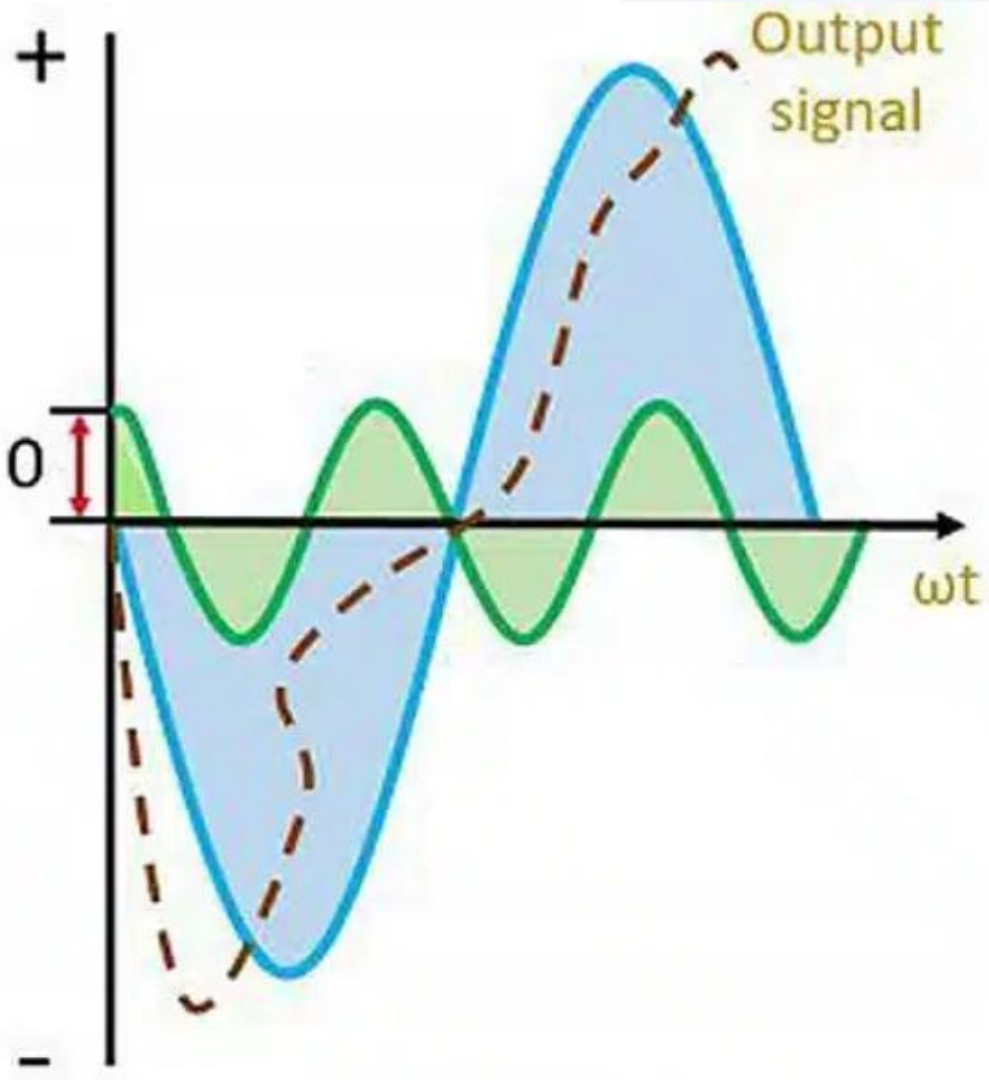
- **Equalization** techniques are used to reduce differences in attenuation across a frequency band.
- For voice-grade telephone lines:
  - Loading coils can change the electrical properties of the line.
  - This helps smooth attenuation effects.
  - Another approach is to use amplifiers that amplify higher frequencies more than lower frequencies.
- **Effect of Equalization**
  - **Without equalization:**
    - Different frequencies experience different attenuation.
    - Signal becomes distorted.
  - **With equalization:**
    - Frequency response becomes flatter.
    - Signal quality improves.

Delay Distortion

- Delay distortion occurs because:
  - The **velocity of propagation** of a signal through a guided medium varies with frequency.
- For a bandlimited signal:
  - Velocity tends to be highest near the center frequency.
  - Velocity decreases toward the edges of the band.
  - Therefore, different frequency components arrive at different times.
- **Result**
  - Different frequency components experience different delays.
  - This causes **phase shifts** between the different frequencies.

Delay Distortion





Delay Distortion and Digital Data

- Consider a sequence of bits being transmitted.
- Because of delay distortion:

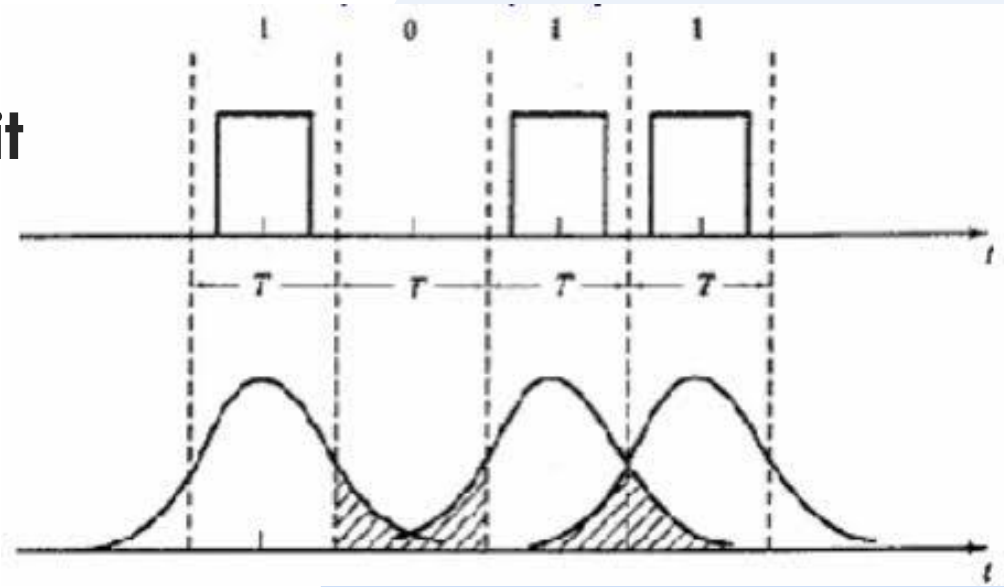
  Signal components belonging to one bit

  ↓

  Spill over into neighboring bit positions

  ↓

  Intersymbol Interference (ISI)



- Therefore:

  Intersymbol interference is a major limitation on the maximum bit rate of a transmission channel.

Noise

- During transmission, the received signal consists of:
  - Transmitted signal
  - Distortion introduced by the transmission system
  - Additional unwanted signals
- The unwanted additional signals are called **noise**.
- Noise is the **major limiting factor in communications system performance**.
- **Four Categories of Noise**
  1. Thermal Noise
  2. Intermodulation Noise
  3. Crosstalk
  4. Impulse Noise

Thermal Noise

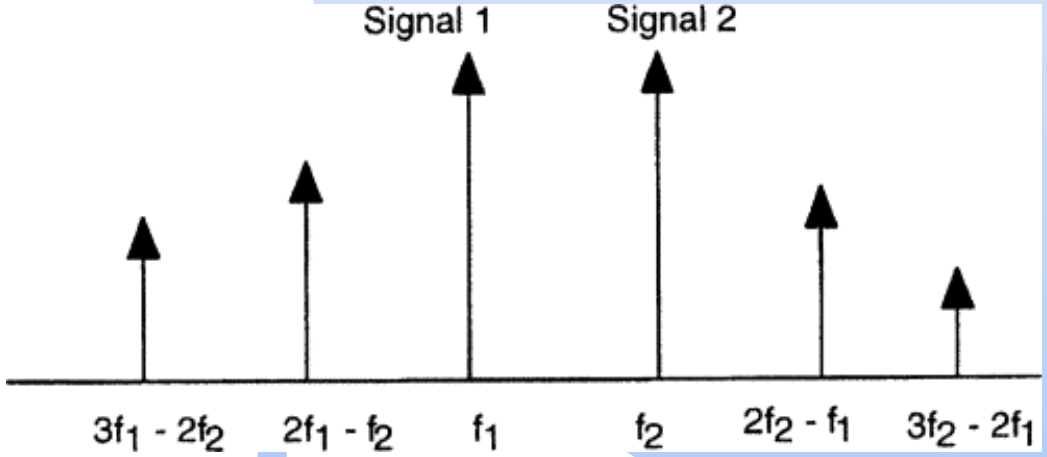
- Thermal noise is caused by:
  - **Thermal agitation of electrons**
- It:
  - Exists in all electronic devices and transmission media.
  - Is a function of temperature.
  - Is approximately uniformly distributed across typical communication bandwidths.
  - Is therefore often called **white noise**.
  - Cannot be eliminated completely.
  - Places an upper bound on communication-system performance.

Intermodulation Noise

- When signals at different frequencies share the same transmission medium, **intermodulation noise** can occur.
- The result can include signals at frequencies that are:
  - The **sum** of the original frequencies
  - The **difference** of the original frequencies
  - Multiples of those frequencies

Cause

- Intermodulation noise is produced by **nonlinearities** in:
  - Transmitter
  - Receiver
  - Transmission medium
- Excessive nonlinearity may result from:
  - Component malfunction
  - Overload caused by excessive signal strength

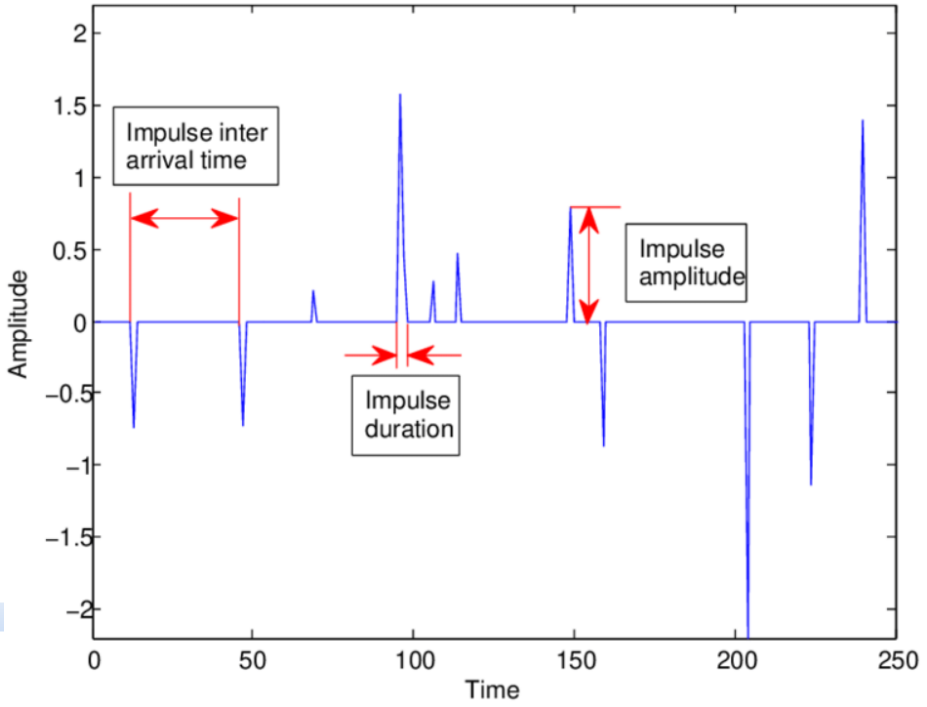


Crosstalk

- **Crosstalk** is unwanted coupling between signal paths.
- **Example**
  - During a telephone conversation, you may hear another conversation from another line.
- Possible causes include:
  - Electrical coupling between nearby twisted pairs.
  - Coupling between coaxial cable lines carrying multiple signals.
  - Unwanted signals picked up by microwave antennas.

Impulse Noise

- Impulse noise is:
  - **Noncontinuous**
  - Made up of irregular pulses or noise spikes
  - Short in duration
  - Relatively high in amplitude
- **Causes:**
  - External electromagnetic disturbances
  - Lightning
  - Faults and flaws in the communication system



Effect of Noise on Digital Data

- The effect of noise can be particularly significant for digital transmission.
- A noise spike can cause:

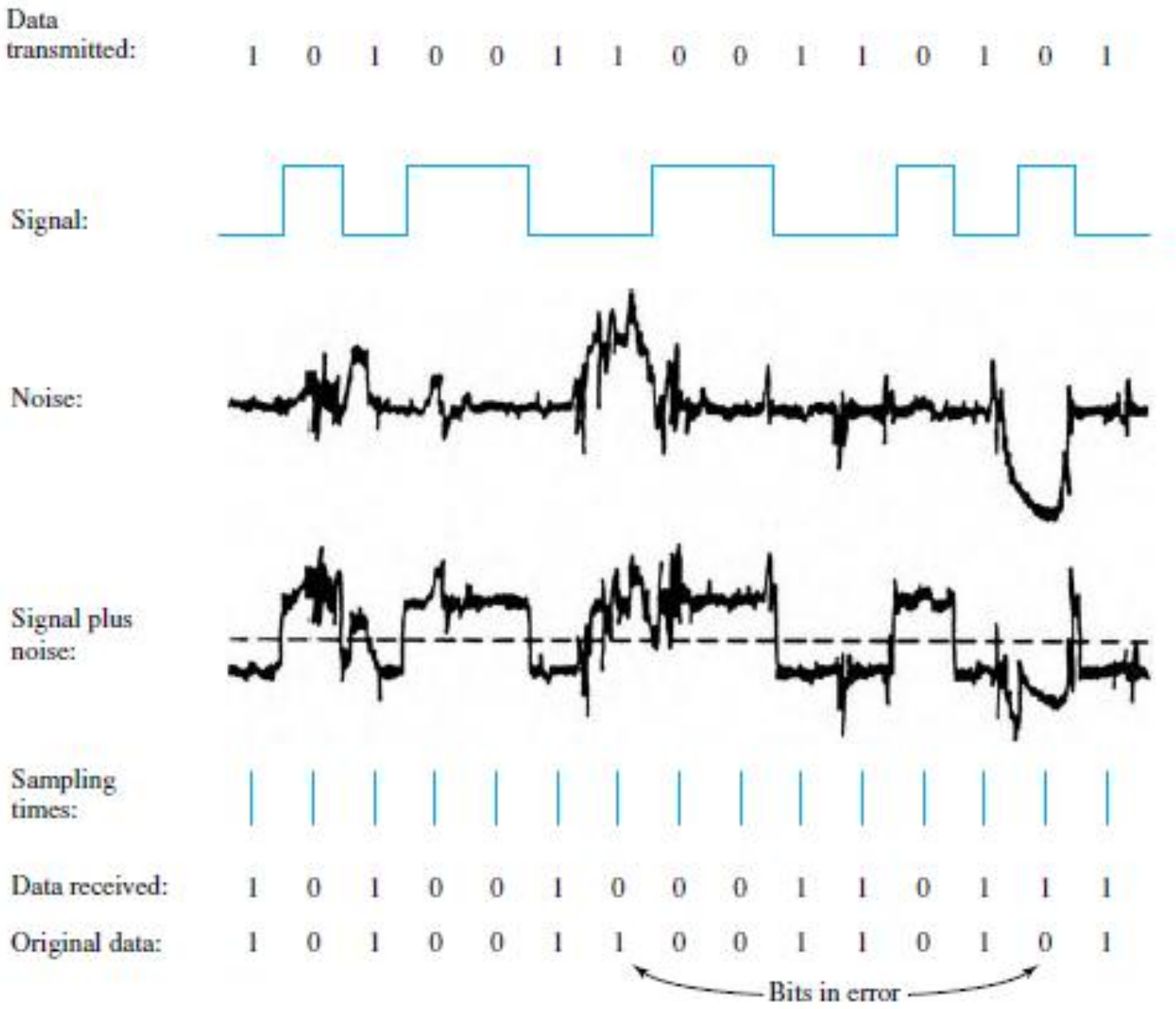
  1 → 0

  or

  0 → 1

- Therefore, the received bit may differ from the transmitted bit.

Effect of Noise on Digital Data



- Image gives an example where a **0.01-second noise spike**:
- Does not destroy voice data in the same way.
- Could affect approximately **560 bits** when digital data are transmitted at **56 kbps**.

Comparing the Four Types of Noise

Type	Basic Cause	Main Characteristic
Thermal noise	Thermal agitation of electrons	Continuous/random noise
Intermodulation noise	Nonlinearities	Produces unwanted frequency components
Crosstalk	Unwanted coupling between signal paths	One signal interferes with another
Impulse noise	External disturbances or system faults	Short-duration, high-amplitude spikes

Effect of Transmission Impairments on Analog vs Digital Signals

- **Analog Signals**
  - Degrade signal quality
  - May affect intelligibility
- **Digital Signals**
  - Transmission impairments:
  - May introduce **bit errors**
    - Example:
      - **Transmitted:** 1
      - **Received:** 0
- **Key Difference**
  - **Analog:** degradation of signal quality
  - **Digital:** possibility of incorrect bits

Review Questions

1. What is the difference between data and a signal?
2. Define analog data and give two examples.
3. Define digital data and give two examples.
4. What is the difference between an analog signal and a digital signal?
5. What are the three parameters of a sine wave?
6. What is bandwidth?
7. What is the difference between an amplifier and a repeater?
8. Why does noise become a problem in analog transmission?
9. Why is digital transmission preferred?
10. Distinguish between simplex, half-duplex and full-duplex transmission.

Thank you

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Lecture 05

DATA TRANSMISSION